Process Biochemistry xxx (2014) xxx–xxx

Contents lists available at ScienceDirect

Process Biochemistry

journal homepage: www.elsevier.com/locate/procbio

Purification and partial characterization of a novel hemagglutinating glycoprotein from the cultured mycelia of *Hericium erinaceus*

Feng-Jie Cui[a,c,*], Yun-Hong Li[a], Xin-Yi Zan[a], Yan Yang[b,**], Wen-Jing Sun[a,c], Jing-Ya Qian[a], Qiang Zhou[c], Si-Lian Yu[c]

[a] School of Food and Biological Engineering, Jiangsu University, Zhenjiang 212013, PR China
[b] National Engineering Research Center of Edible Fungi, Shanghai Academy of Agricultural Sciences, Shanghai 201403, PR China
[c] Jiangxi Provincial Engineering and Technology Center for Food Additives Bio-production, Dexing 334221, PR China

ARTICLE INFO

Article history:
Received 2 February 2014
Received in revised form 7 April 2014
Accepted 9 April 2014
Available online xxx

Keywords:
Hericium erinaceus, Cultured mycelia
Glycoprotein
Hemagglutinating activity
Structural novelty

ABSTRACT

HEG-5, a novel glycoprotein with hemagglutinating activity, was firstly isolated and purified from the cultured mycelia of *Hericium erinaceus* CZ-2. SDS–PAGE, Native-PAGE and MALDI-TOF-MS proved that HEG-5 was a single band with the molecular weight of approximately 14.4 kDa. HEG-5 had the protein: polysaccharide ratio of approximately 10:1 (%/%) and contained D-glucose, L-rhamnose, D-galactose and D-mannose with a molar ratio of 1.00:1.09:2.45:7.14 in polysaccharide fraction. HEG-5 was an acidic glycoprotein with a PI value of 6.3 and the higher content of acidic amino acids (Asp, 12.42 ± 0.25% and Glu, 12.24 ± 0.26%) in protein fraction. FT-IR and NMR spectra revealed that HEG-5 contained the protein and carbohydrate portions with (1→4)-linked β-galactose residues and β-linked glucose residues. Circular dichroism (CD) demonstrated that HEG-5 was a β-sheet predominant glycoprotein. Hemagglutination assay proved it was a thermo-unstable glycoprotein. The HEG-5 structural novelty was finally presented by protein sequencing and modeling by using MALDI-TOF-MS, NCBI blast search and online SWISS-MODEL Workspace service.

© 2014 Elsevier Ltd. All rights reserved.

1. Introduction

Hericium erinaceus, an edible mushroom belonging to the Aphyllophorales, Hydnaecae, is a well known mushroom in Asia due to its nutrition and effectiveness in treating gastric and duodenal ulcers [1]. Previous studies also have demonstrated that *H. erinaceus* possesses the anti-tumor [2], anti-obesity [3], antioxidant [4], anti-platelet [5], and antimicrobial [6] activities. Recently, many valuable bioactive compounds such as polysaccharides [7], dilinoleoyl-phosphatidylethanolamine [8], amycenone [9], aromatic compounds [10], and D-arabinitol ester [11] have been isolated from *H. erinaceus*. Among them, polysaccharides have attracted considerable attentions. For example, the heteropolysaccharide (HEPF1), with a (1→6)-linked α-D-galactopyranosyl backbone and the α-D-glucan (HEPF5) composed of a multi-branch tetrasaccharide repeating unit had been purified from the fruiting bodies of *H. erinaceus* [7,12].

Glycoprotein and/or glycopeptide have different structures with polysaccharides due to their protein/polypeptide chains bound covalently and specifically with carbohydrate side-chains [13,14]. β-glucan-protein, α-glucan-protein and heteroglycan-protein complexes in mushrooms are found to possess the antitumor activities [15]. Until now, only two glycoproteins HEL and HEA have been obtained from fruiting bodies of *H. erinaceus*. HEL is a sialic acid-binding lectin purified from the fresh fruiting body of *H. erinaceum* with a blocked N-terminal and a molecular mass of approximately 54 kDa [16]. HEA from the dried *Hericium erinaceum* fruiting body has the molecular mass of about 51 kDa with the novel N-terminal amino acid sequences. It also possessed the hemagglutinating activity and antiproliferative activity toward HepG-2 and MCF-7 cells [17]. Our research group previously screened and optimized the submerged culture parameters to maximize the submerged-cultured *H. erinaceus* mycelia production by using statistical experiment design [18,19]. As for the mushroom *H. erinaceus*, isolating and structurally elucidating the glycoproteins from the cultured mycelia are still necessary.

The present study aimed to: (1) isolate and purify the glycoproteins from cultured mycelia of *H. erinaceus* with antitumor

* Corresponding author at: School of Food and Biological Engineering, Jiangsu University, Zhenjiang 212013, PR China. Tel.: +86 511 88780226; fax: +86 511 88780201.
** Corresponding author. Tel.: +86 511 88780226; fax: +86 511 88780201.
E-mail addresses: fengjiecui@163.com, fengjiecui@mail.ujs.edu.cn (F.-J. Cui), yangyan@saas.sh.cn (Y. Yang).

http://dx.doi.org/10.1016/j.procbio.2014.04.008
1359-5113/© 2014 Elsevier Ltd. All rights reserved.

and/or hemagglutinating activities, (2) evaluate its primary thermostability under the 30–100 °C treatment, and (3) elucidate the structural characteristics and predict the potential 3D structure by using FT-IR, Circular dichroism (CD), NMR spectroscopy and MALDI-TOF-MS.

2. Materials and methods

2.1. Strain and culture conditions

H. erinaceus CZ-2, previously isolated by our group from the mountainous district of Anhui Province in China, was used in the present study [18,19]. The stock culture was maintained on potato-dextrose-agar (PDA) slant at 4 °C, and sub-cultured every two months. The seed culture was obtained by inoculating the stock culture (about 10 pieces) into the medium containing corn flour, 30.85 g/L; yeast extract, 2.81 g/L; corn steep liquor, 16.9 mL/L; glucose, 10 g/L; KH_2PO_4, 1 g/L; and $MgSO_4 \cdot 7H_2O$, 0.5 g/L, in a 250 mL Erlenmeyer flask, at 25 °C for 7 d, and then transferred to a 20-L mechanical stirred fermentor (GRCB, Green Bio-engineering Co., Ltd., Zhenjiang, China) with working volume of 12 L with an air flow volume 0.8 v/min with agitation speed of 100 rpm at 25 °C for fermenting the *H. erinaceus* mycelia. After 7-d fermentation, mycelia were filtered off, washed with distilled water, and kept in frozen condition.

2.2. Isolation and purification of glycoprotein HEG-5

The *H. erinaceus* mycelia (500 g) was blended with distilled water (400 mL) for 1 min and then continuously stirred for 2 h to obtain homogeneous slurry. The slurry was stored at 4 °C for 24 h and centrifuged at 5030 × *g* for 20 min to collect the supernatant. The supernatant was precipitated with 80% saturated ammonium sulphate overnight at 4 °C. The precipitate collected by centrifugation (12,864 × *g*, 20 min) was dissolved and dialyzed against distilled water to obtain the crude mushroom extract. Then the crude extract was applied to a column (2.5 cm × 25 cm) DEAE-Sepharose Fast Flow anionic resin (GE Healthcare, Uppsala, Sweden), with the sequential elution of 0.1, 0.3 and 0.5 M NaCl at pH 7.0 (every fraction tube 8 mL). The carbohydrate content of the eluates was determined by measuring the absorption at 490 nm using the phenol–sulfuric acid method. The protein concentration was determined spectrophotometrically at 280 nm. Three factions HEG-1, HEG-3, and HEG-5 were collected, dialyzed and freeze-dried. The fraction HEG-5 was proven to have the highest antitumor activity by comparing the tumor cell growth inhibition ratio of each fraction. HEG-5 showed a single peak eluting with distilled water (pH 7.0) on a Superdex™ 75 prep grad column (GE Healthcare, Uppsala, Sweden) using the AKTA purifier 100 (GE Healthcare, Uppsala, Sweden).

2.3. Molecular mass determination using SDS–PAGE and MALDI-TOF/TOF

Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) was used to check the purity and primary molecular weight of HEG-5 by dissolving the sample in the distilled water and mixing with SDS–PAGE sample loading buffer containing DTT to reduce the protein. Then the HEG-5 molecular weight was obtained by comparing and calculating its electrophoretic mobility with those of standard molecular weight markers.

The molecular weight of HEG-5 was also determined by using an AB 5800 MALDI-TOF-MS (AB SCIEX, MA, USA) in linear mode using 20-kV ion acceleration without post-acceleration. The spectrum was recorded at a detector voltage of 1.65-kV, and was expressed as the averaged results of at least 300 laser shots. The matrix compound was gentisic acid DHB (2,5-dihydroxy benzoic acid). HEG-5 fraction was dissolved in 0.1% TFA: acetonitrile (2:1 v/v) and mixed with the matrix solution (1:4 v/v). A mass calibration procedure was employed prior to the analysis of a sample using protein calibration standards I (Brucker Daltonics, IL, USA).

2.4. Native-PAGE and isoelectric focusing

Native-PAGE of HEG-5 was carried out on ice using a 12% resolving gel and a 5% stacking gel. The gel was stained with Coomassie Brilliant Blue. Isoelectric focusing (Bio-Rad Gel IEF, pH 3–7) was done in a Phastsystem (GE Healthcare, Uppsala, Sweden). Gels were stained with Coomassie Brilliant Blue.

2.5. Carbohydrate content and monosaccharide compositions

Carbohydrate content of HEG-5 was determined by the phenol–sulfuric acid method using glucose as the standard. The monosaccharide composition analysis of HEG-5 was conducted by using absolute hydrolysis according to the previously described method [20] with some modifications. Thirty mg of HEG-5 was mixed with 0.3 mL of 72% sulfuric acid in a sealed pressure tube and incubated at 30 °C for 60 min. Dilute the concentration of sulfuric acid to 4% by adding 8.4 mL de-ionized water and autoclave for one hour at 121 °C. The hydrolysates was filtered and applied for the monosaccharide composition in a 6890 N GC (Agilent Technologies, Santa Clara, CA, USA) with capillary column (30 m × 0.32 mm) and a flame-ionization, using L-rhamnose, D-arabinose, D-mannose, D-galactose and D-glucose as the standards.

2.6. Protein content and amino acid composition

Protein content was analyzed by Micro BCA Protein Assay Reagent Kit (Labaide Biotechnology, Shanghai, China) using bovine serum albumin as the standard. Amino acid composition of HEG-5 was determined by hydrolysis under vacuum in 6 M HCl at 110 °C for 24 h and analysis with a Hitachi 835-50G automatic amino acid analyzer (Hitachi, Tokyo, Japan).

2.7. Conformation analysis

FT-IR spectroscopic analysis of HEG-5 was carried out by using a Nicolet Fourier transform infrared spectrometer (Thermo Nicolet, MA, USA) with a scanning range of 400–4000 cm^{-1}. About 1 mg of freeze-dried HEG-5 was mixed with KBr in 1:100 ratio and finally pressed to be a pellet about 1 mm thick. ^1H NMR spectra of HEG-5 were done by a Brucker DRX 500 MHz (Brucker, IL, USA). Samples were dissolved in D_2O and mainly examined at room temperature. Chemical shifts were referenced to residual HDO at d 4.78 ppm as the internal standard. Circular dichroism (CD) spectra of HEG-5 (1.0 mg/mL, in distilled water) was obtained on a JASCO J-810 spectropolarimeter (Jasco, Tokyo, Japan) at 25 °C in the region of 190–250 nm with a scanning speed of 100 nm/min. The percentages of α-helix, β-sheet and β-turn were calculated by de-convoluting CD analysis using CD analysis program, CONTIN, SELCON3 and CDSSTR. All of the three programs were used with a reference protein set consisting of 48 proteins.

2.8. In-gel tryptic digestion

The band of HEG-5 in SDS–PAGE was excised, rinsed three times with double distilled water, and destained with 200 μL destaining solution (25 mM NH_4HCO_3, 50% CH_3CN). The gel slices were vacuum-dried, and added to a trypsin solution to incubate at 37 °C overnight. The supernatant was collected when the reaction was cooled to 20 °C. The gel was subsequently extracted twice with 20 μL 0.1% TFA in 50% ACN by gently mixing and incubation at 60 °C for 10 min. The supernatant and extract solution were combined and freeze-dried for further MALDI analysis.

2.9. Mass spectrometry and protein identification

MALDI mass spectrometric analysis was performed on AB 5800 MALDI-TOF (AB SCIEX, MA, USA). Spectra were accumulated until a satisfactory S/N had been obtained. Parent mass peaks with the range from 400 to 1600 *m/z* were picked out for MS/MS analysis. The peptide mass finger-printings of glycoprotein and MS/MS data were acquired and processed using Mass-Lynx V4.1 software (Waters, MA, USA) and were converted to PKL files by the Protein-Lynx 2.2.5 software (Waters, MA, USA). The PKL files were analyzed using the MASCOT search engine. The search parameters were defined as follows: database, NCBInr; taxonomy, Homo sapiens; enzyme, trypsin; and allowance of one missed cleavage.

2.10. Protein modeling

Automatic modeling mode of the SWISS-MODEL Workspace was used to build 3D structure model of HEG-5. For template selection, the template's quality has been predicted from features of the target-template alignment and the templates with the highest quality have then been selected for model building. Models were built on the basis of target-template alignment. Coordinates which were conserved between the target and the template were copied from the template to the model.

2.11. MTT cell viability assay

Gastric cancer cell line SGC-7901 cells purchased from American Tissue Culture Collection (ATCC, MD, USA) were cultured in 5% CO_2 at 37 °C in RPMI 1640 medium containing polymyxin B (PMB) and 10% fetal calf serum that was supplemented with 100 U/mL penicillin and 100 μg/mL streptomycin. The anti-tumor activity of the *H. erinaceus* glycoprotein fractions was determined as follows. Cells (2×10^4/well) in their exponential growth phase were seeded into each well of a 96-well flat-bottomed culture plate and incubated for 24 h before addition of different fractions of *H. erinaceus* with the concentration of 25, 50, 100, 200 μg/mL. Cells were incubated for another 48 h, then 20 μL of the combined MTT (5 mg/mL)/PBS solution was added into each well of the 96 well assay plate. Cells were incubated at 37 °C for 4 h and the formazan product was solubilized with dimethylsulfoxide (DMSO). The absorbance was detected in the microplate reader (BioTek ELX800, VT, USA) at 490 nm, and measured values were expressed as mean ± SD.

2.12. Hemagglutinating and hemolytic assay

HEG-5 was dissolved in phosphate-buffered saline (PBS, pH 7.2) and diluted in two-fold series. Each 25 μL aliquot of the sample solution was pipetted into a microtitre V-plate and added with 25 μL of the rat erythrocytes suspension (2%; v/v). The agglutination process kept for 1 h. The hemagglutination titer (one hemagglutinating unit) of the samples was expressed as the reciprocal of the highest dilution exhibiting visible agglutination of rat erythrocytes. Specific activity was the number of hemagglutination units per mg protein.

Please cite this article in press as: Cui F-J, et al. Purification and partial characterization of a novel hemagglutinating glycoprotein from the cultured mycelia of *Hericium erinaceus*. Process Biochem (2014), http://dx.doi.org/10.1016/j.procbio.2014.04.008

To investigate the hemagglutinating inhibition of various sugars including D-glucose, D-mannose, D-galactose, D-xylose, D-fructose, L-rhamnose, maltose, inositol, L-sorbose, L-arabinose, D-arabinose, D-melibiose, sucrose, D-sorbitol, D-mannitol, D-cellobiose and α-lactose, serial 2-fold dilutions of sugar samples were prepared in PBS (pH 7.2). All of the dilutions were mixed with an equal volume (25 μL) of a solution of the glycoprotein sample with 2 mg/mL. The mixture was kept for 30 min at room temperature and then mixed with 25 μL of a 2% rat erythrocyte suspension. The hemagglutination titers obtained were compared with a non-sugar containing blank.

The thermostability of HEG-5 was determined by comparing the agglutinating activity after treated at the temperatures ranging from 30 to 100 °C for 30 min. Circular dichroism was also used for determining the structure changes after the treatment of HEG-5 at 30, 50, 70, 100 °C.

Hemolytic assay was conducted according to the described method [21] with some modifications. HEG-5 was diluted to a series sample solution (0.2 mL) with various concentrations ranging from 62.5 μg/mL to 10 mg/mL, mixed with 2% (v/v) rat erythrocytes (0.2 mL), and incubated at room temperature for 1 h before centrifugation at 1000 × g for 5 min. Additions of 0.2 mL PBS and 0.1% (w/v) Triton X-100 to the 0.2 mL 2% (v/v) rat erythrocytes were used as control and positive groups, respectively. The amount of hemoglobin released from disrupted erythrocytes was determined spectrophotometrically at 540 nm. One hundred percent hemolysis was defined as OD_{540} of hemoglobin released from erythrocytes treated with 0.1% Triton X-100.

2.13. Statistical analysis

Each experiment was repeated three times using duplicate samples. The results were expressed as means ± standard deviations. Statistical comparisons were made by one-way analysis of variance (ANOVA), followed by Duncan's multiple-comparison test. Differences were considered significant when the p-values were <0.05.

3. Results and discussion

3.1. Isolation and purification of the glycoprotein

The crude extract from *H. erinaceus* at the concentration of 200 μg/mL exhibited a growth inhibition rate of 47.5% on SGC-7901 cells during 48-h incubation. The crude extract was applied to DEAE-Sepharose Fast Flow column and three fractions (named HEG-1, -3 and -5) were recovered (Fig. 1(A)). The growth inhibition rate of each fraction was shown in 4(A). At the highest concentration of 200 μg/mL, the inhibition ratios of HEG-1, -3 and -5 on SGC-7901 cells reached to 56.5%, 68.7% and 93.4% in 48 h, respectively. Among them, the fraction HEG-5 had the highest growth inhibitory effect.

3.2. Molecular mass estimation of HEG-5

SDS–PAGE followed by coomassie brilliant blue staining revealed that HEG-5 was a single band corresponding to a

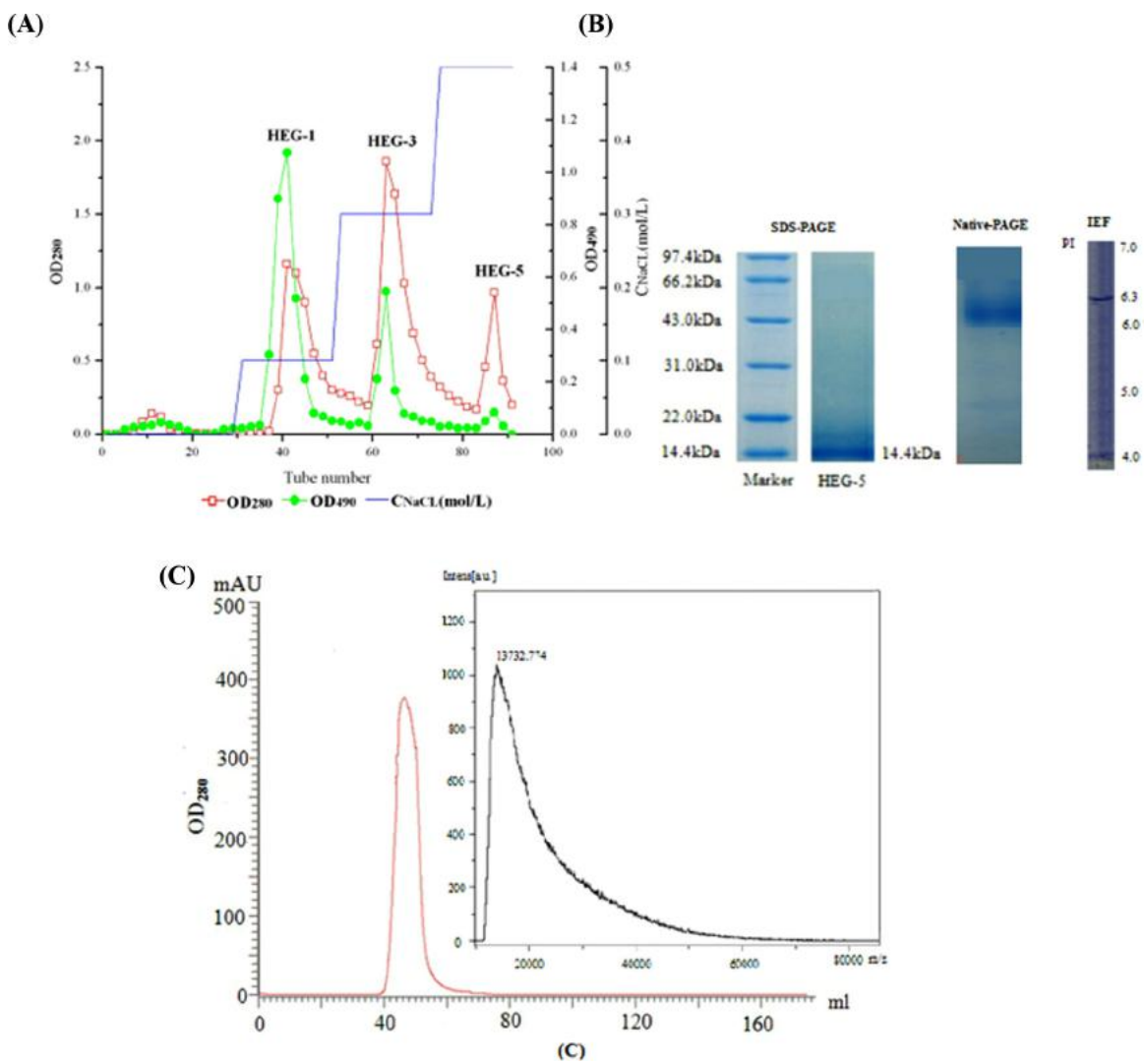

Fig. 1. (A) Anion-exchange chromatography of crude extract from *H. erinaceus* CZ-2 on DEAE-Sepharose Fast Flow column (2.5 cm × 20 cm). (B) SDS–PAGE Native-PAGE and IEF of HEG-5. The markers of SDS–PAGE were phosphorylase b (97.4 kDa), bovine serum albumin (66.2 kDa), ovalbumin (43.0 kDa), carbonic anhydrase (31.0 kDa), soybean trypsin inhibitor (22.0 kDa), and α-lactalbumin (14.4 kDa), all from GE Healthcare.

Please cite this article in press as: Cui F-J, et al. Purification and partial characterization of a novel hemagglutinating glycoprotein from the cultured mycelia of *Hericium erinaceus*. Process Biochem (2014), http://dx.doi.org/10.1016/j.procbio.2014.04.008

molecular mass of 14.4 kDa (Fig. 1(B)). MALDI-TOF-MS spectra also have been used as an effective method to give the more precise molecular weight. For example, by using MALDI-TOF-MS, the molecular masses of glycoeproteins PPL from the mushroom *Pleurocybella porrigens* and POL from the mushroom *Pleurotus ostreatus* were predicted as 56 kDa and 81.6 kDa, respectively [22,23]. In the present study, MALDI-TOF-MS spectra of HEG-5 gave a single peak at *m*/*z* 13 733 (Fig. 1(C)), which agreed with the molecular mass of 14.4 kDa in SDS–PAGE.

3.3. Native-PAGE and isoelectric focusing

The HEG-5 Native-PAGE also was a single band indicating its high purity possibly with no subunits (Fig. 1(B)). The isoelectric focusing of HEG-5 on polyacrylamide gel in the pH range 4–7 revealed a single band at pH 6.3 as the PI (Fig. 1B), proved that HEG-5 was an acidic protein, similar to the previously purified *Pleurotus citrinopileatus* glycoprotein PCP-3A with the PI of 6.85 [24].

3.4. Carbohydrate content and monosaccharide compositions

The carbohydrate content of the glycoprotein HEG-5 was found to be 8.86 ± 0.54% by the phenol–sulfuric acid method. The monosaccharide compositions of HEG-5 were determined as D-glucose, L-rhamnose, D-galactose and D-mannose with a molar ratio of 1.00:1.09:2.45:7.14 by analyzing their alditol acetate derivatives. D-mannose was the most abundant monosaccharide with 61.125% (mol/mol) of the total monosaccharides, which has the difference with other mushroom glycoproteins. For instance, galactose was the monosaccharide with the highest content of 46.73% (mol/mol) in HM-3A from *Hypsizygus marmoreus* fresh fruiting body [25].

3.5. Protein content and amino acid composition

The BCA assay showed that HEG-5 contains 86.67 ± 2.79% of soluble protein portion, which proven that HEG-5 was protein-dominate. The result is similar with PLG-3 from the mycelia of *Phellinus linteus* AML1101, which contains 85.5% of protein portion [26]. Amino acid analysis results (Table 1) revealed that HEG-5 contained 17 kinds of amino acids composed of a high content of acidic amino acid (Asp, 12.42 ± 0.25% and Glu, 12.24 ± 0.26%) and a low content of Met (2.04 ± 0.04%) and Cys (1.15 ± 0.03%), which agreed with the conclusion of the acidic glycoprotein of HEG-5 from isoelectric focusing result. Generally, the high contents of acidic amino acids mean that the glycoprotein/protein should be acidic. The glycoprotein PPL with 21.7% (mol%) acidic amino acids was proven as an acidic glycoprotein with a PI of 5.7 [22].

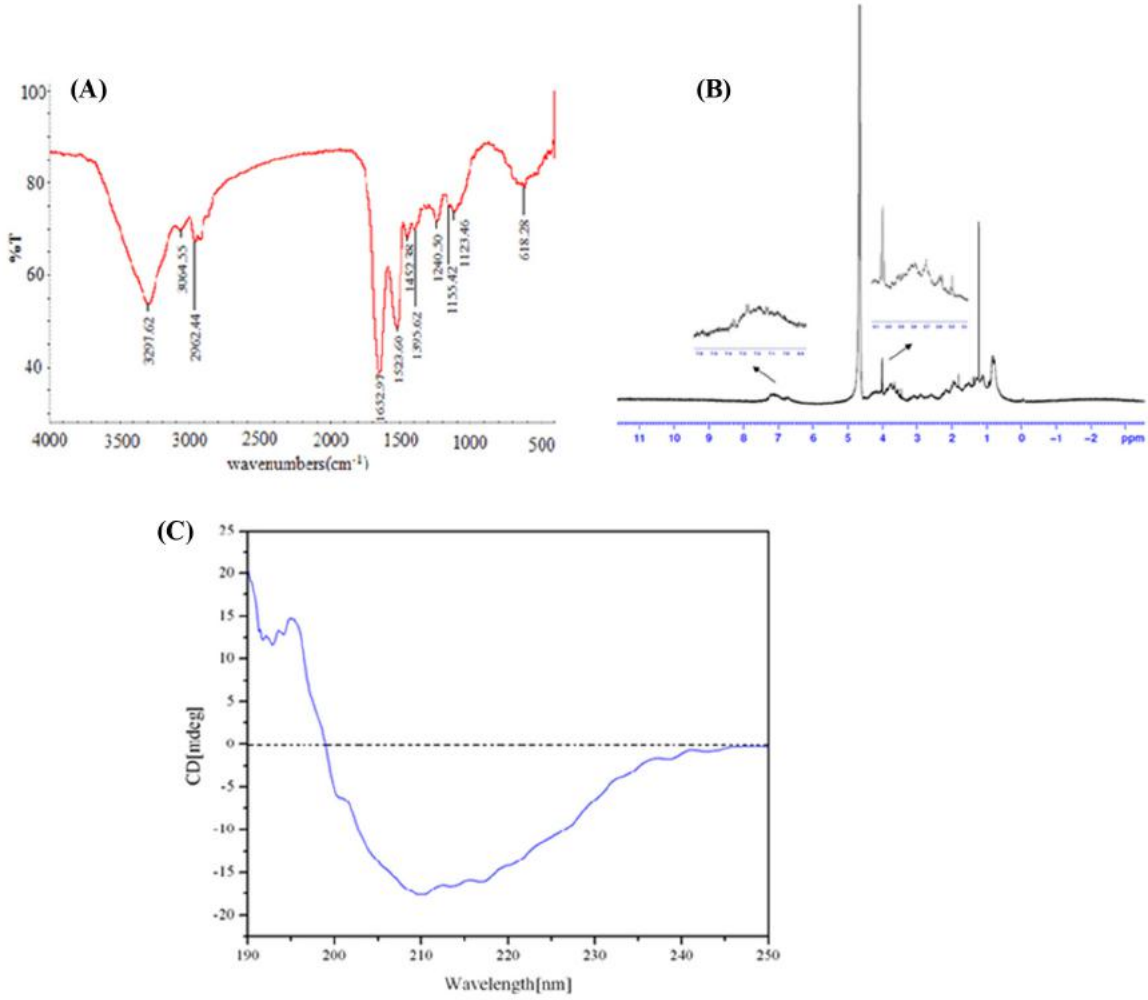

Fig. 2. (A) Fourier transform infrared (FT-IR) spectra of glycoprotein HEG-5. (B) ^{1}H NMR spectrum of glycoprotein HEG-5 in D_2O. (C) Circular dichroism (CD) spectrum of glycoprotein HEG-5.

Table 1
Amino acid composition of HEG-5.

Amino acid	Mass%	Amino acid	Mass%
Asp	12.42 ± 0.25	Cys	1.15 ± 0.03
Glu	12.24 ± 0.26	Val	7.17 ± 0.18
Ser	5.56 ± 0.13	Met	2.04 ± 0.04
His	2.72 ± 0.04	Phe	5.58 ± 0.12
Gly	6.21 ± 0.15	Ile	5.49 ± 0.13
Thr	6.07 ± 0.12	Leu	7.38 ± 0.17
Arg	4.81 ± 0.07	Lys	4.76 ± 0.05
Ala	6.84 ± 0.16	Pro	4.17 ± 0.05
Tyr	5.39 ± 0.13	Trp	n.d.[a]

[a] Not determined.

3.6. Conformation analysis

Fig. 2A presents the diffuse reflectance Fourier transform infrared (FT-IR) spectra of glycoprotein HEG-5. The broad stretching peak around 3297 cm^{-1} was ascribed to the hydroxyl groups of polysaccharides and N-H stretching vibrations of protein. The weak band around 2900 cm^{-1} was attributed to C—H stretching vibrations and bending vibrations. The bands at 1652 cm^{-1} and 1523 cm^{-1} was assigned to the absorbance of the C=O stretching vibrations and N—H bending vibrations of the amide groups, suggesting that HEG-5 contains α- and β-configurations of protein. Absorption bands at 1452 cm^{-1} and 1395 cm^{-1} was C—N stretching vibrations and may be due to carboxylate groups of galacturonic acid residues. The band at 1240 cm^{-1} was assigned to the stretching vibration of C=O groups. The wave numbers 1155 cm^{-1} and 1123 cm^{-1} was due to C—N stretching vibrations. The band around 618 cm^{-1} was attributed to N—H bending vibrations of the amide groups.

The chemical structure of HEG-5 was also studied by ^1H NMR spectroscopy (Fig. 2B). The signals of ^1H NMR at 7.3–7.1 ppm were assigned to the benzene ring of phenylalanine residues. Signals of ^1H NMR at 4.7 ppm were attributed to the proton of water (H_2O) contained in the solvent (D_2O). The peaks at 4.0 ppm may be caused by protons of heteroatoms O, N in compounds AlkOH, ArOH, $ArNH_2$, ArNHR. Signals of ^1H NMR at 3.7 and 3.8 ppm were assigned to H-2 and H-3 of (1→4)-linked β-galactose residues, respectively. The signals at 3.6–3.5 ppm represents H-2 of the β-linked galactose and 3.4–3.3 ppm were assigned to the H-2 of the β-linked glucose. The proton NMR for HEG-5 also showed that a significant amount of acetyl groups in the region 2.0–1.8 ppm. The peaks at 1.4–1.2 ppm represented the methyl group of rhamnose residues.

Based on CD data, the secondary structure content of HEG-5 was estimated to be 13.9% of α-helix, 56.5% of β-sheet and 29.6% of random-coil structure, indicating HEG-5 was a β-sheet predominant glycoprotein with a relatively low α-helical content (Fig. 2C). However, secondary structure of CML (*Cordyceps militaris* lectin) was reported to contain a relatively higher content of α-helix than β-sheet. It contained 27% of α-helix, 12% of β-sheet, 29% of β-turn and 32% of random-coil structure [27].

3.7. Mass spectrometry and protein identification

The sequence of HEG-5 was analyzed to search for homologous proteins in the currently known databases. The results of mass spectrometry revealed that the HEG-5 sequence is similar to peptidase M 23 [*Halomonas* sp. HAL1] (gi|496245751) of

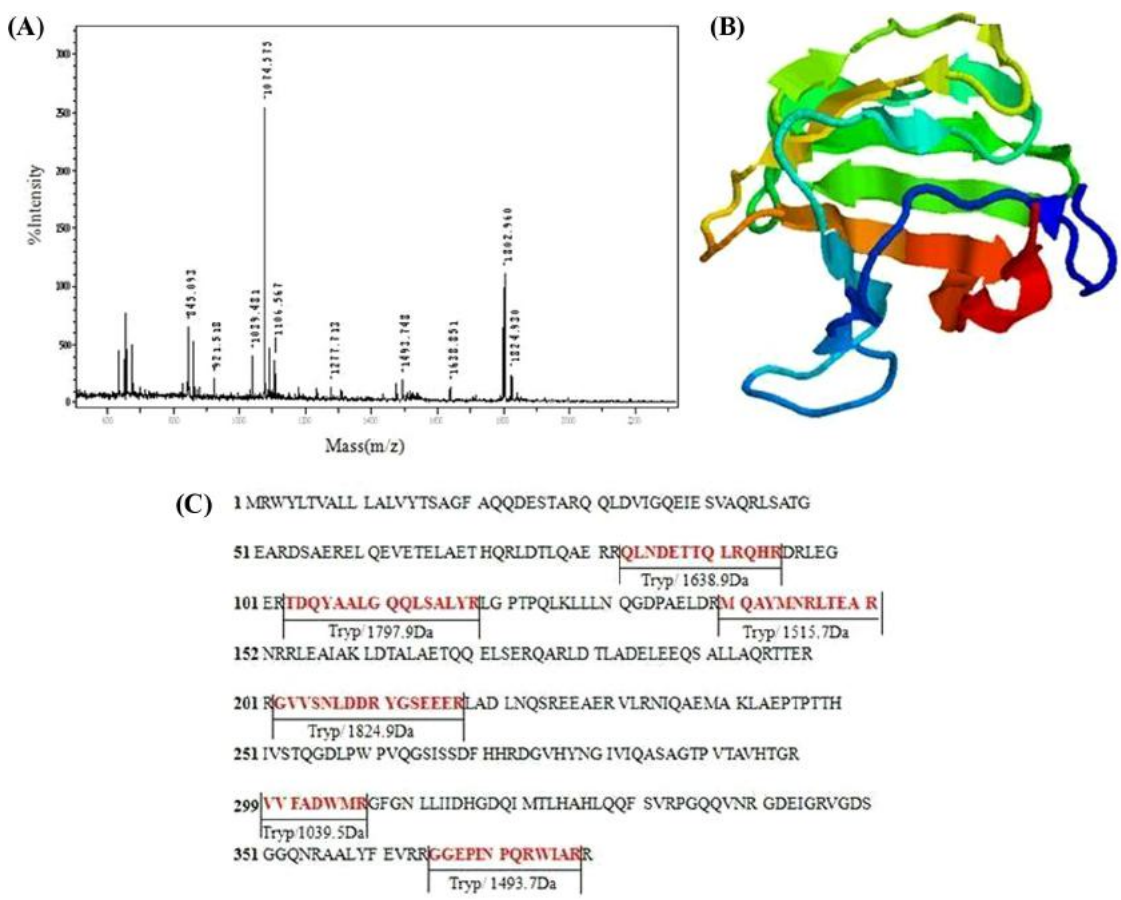

Fig. 3. (A) Mass spectra of peptides of tryptic digested HEG-5 by MALDI-TOF-MS. (B) The predicted three-dimensional structure of HEG-5 obtained by online SWISS-MODEL Workspace. (C) Peptide Mass Fragments of HEG-5 confirmed by a NCBI blast search of mass spectrometry.

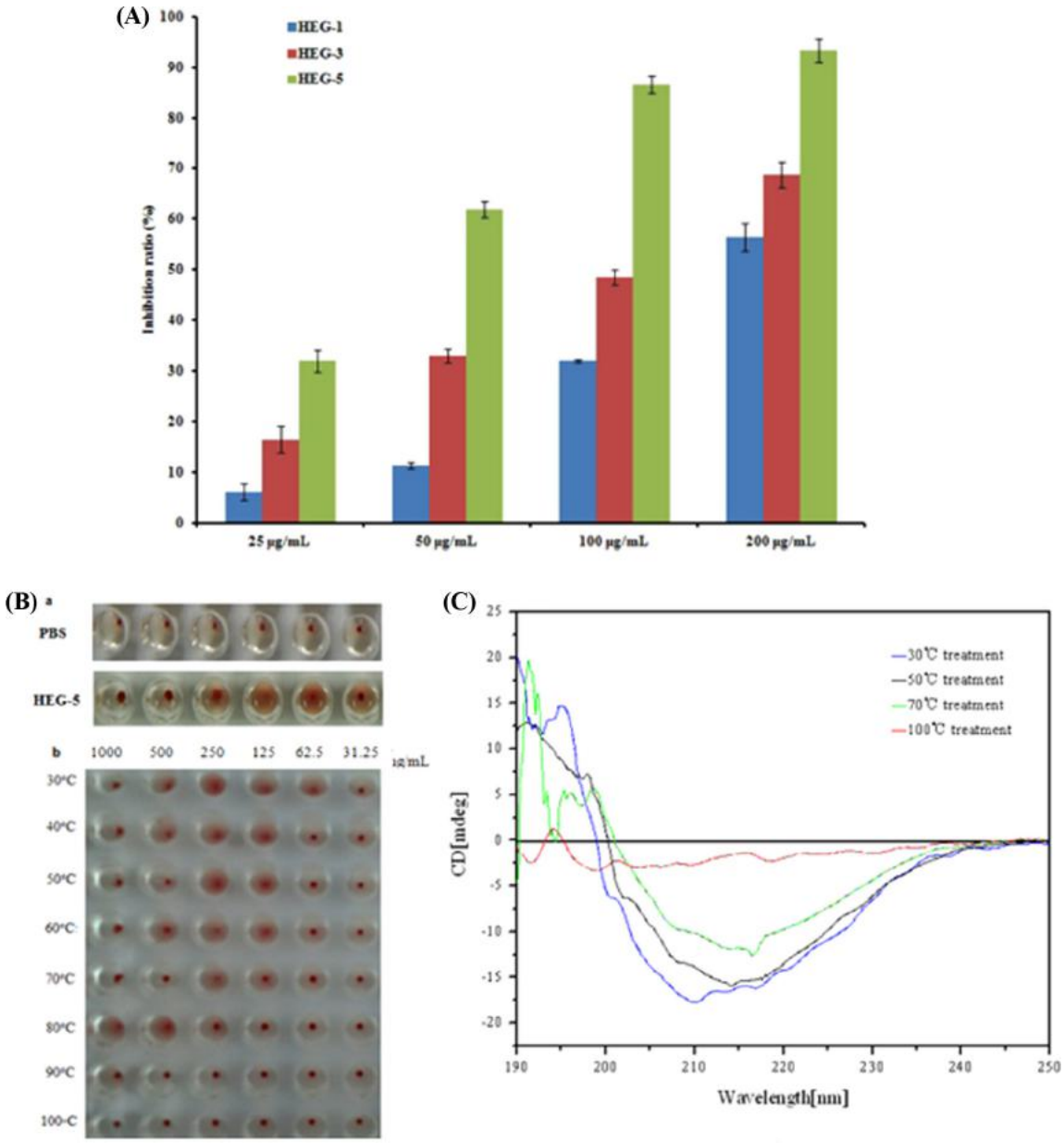

Fig. 4. (A) The growth Inhibitory effect of proliferation of SGC-7901 cells (2×10^5 cells/mL) incubated with different concentrations of glycoprotein fractions from the cultured mycelia of *H. erinaceus* CZ-2 at 37 °C for 48 h. (B) (a) Hemagglutinating activity of HEG-5 on rat erythrocytes. (b) Effect of temperature on hemagglutinating activity of HEG-5. (C) Thermostability of HEG-5 identified by Circular dichroism (CD) spectrum.

42,630 Da confirmed by mass spectrometry (MADIF-TOF-MS) and an NCBI BLAST search (Fig. 3A and C). However, the results of MS/MS revealed that it possessed the similar amino acid sequence to the hypothetical protein [Bacillus cereus] gi|446583541 of 15754 Da (data not shown). All above results prove that HEG-5 was a novel glycoprotein.

3.8. Protein modeling

The online SWISS-MODEL Workspace was used to predict the HEG-5 3D structure. The overall fold of the HEG-5 model was shown in Fig. 3B. It was noteworthy that β-sheets occurred predominantly in the structures of HEG-5, which is confirmed by the CD data. From the model, HEG-5 had the not tightly folded 3D structure. As discussed previously, this method for predicting the glycoprotein 3D structure has its limit due to the incomplete sequence information and lower identity to the known protein [14]. The specifically bound polysaccharide chain is also an obstacle to obtain the accurate structure information. Hence, further investigations including crystallization are necessary to provide more structural information of HEG-5.

3.9. Antiproliferative activity of HEG-5

Fig. 4(A) showed the growth inhibitory effect of HEG-5 against SGC-7901 cells at various concentrations. The effect increased with HEG-5 concentration in the range of 25–200 μg/mL. The growth inhibition rate of HEG-5 at the concentration of 200 μg/mL reached $93.42 \pm 2.63\%$ in 48 h. Recently, other mushroom glycoproteins also have been regarded as the antitumor agents with the inhibitory activity against various cancer cells. For instance, PCP-3A from *Pleurotus citrinopileatus* inhibited the growth of U937 cells by 97.70% in

Table 2
Effect of temperature on hemagglutinating activity of HEG-5.

Temperature (°C)	30	40	50	60	70	80	90	100
Hemagglutinating activity remained [U][a]	16	8	8	8	4	2	0	0

[a] Initial hemagglutinating activity was 16 U.

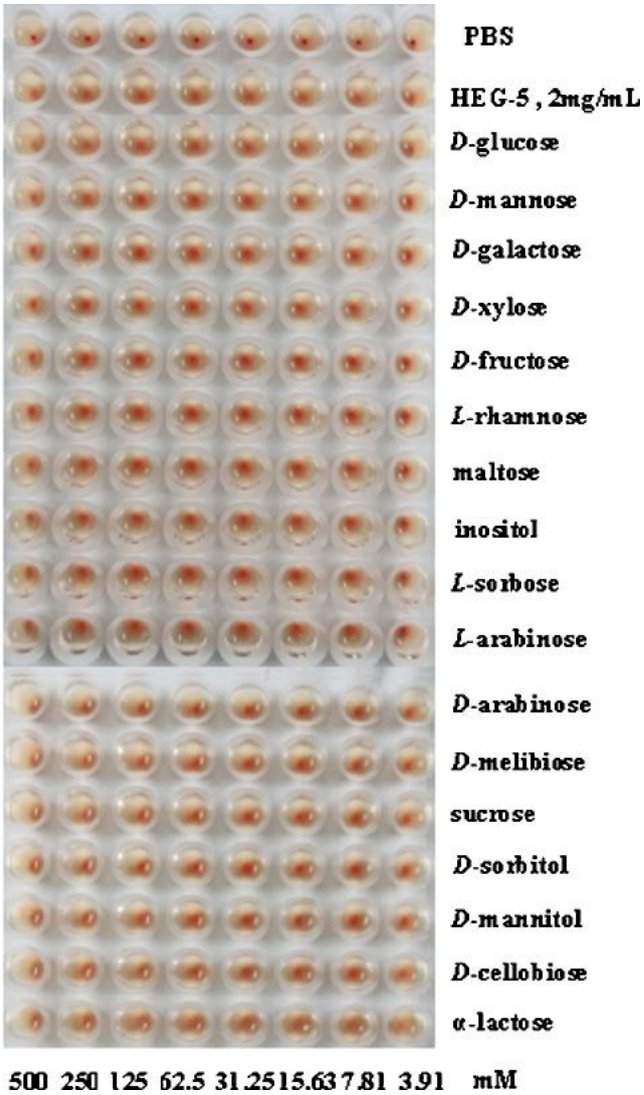

Fig. 5. Hemagglutinating inhibition tests by various sugars. PBS: 25 μL 2% rat erythrocytes suspension mixed with 50 μL PBS; HEG-5, 2 mg/mL: 25 μL 2% rat erythrocytes suspension mixed with 50 μL 2 mg/mL HEG-5; sugars: 25 μL 2% rat erythrocytes suspension mixed with 25 μL 2 mg/mL HEG-5 and 25 μL various concentrations of different sugars.

72 h with the concentration of 50 μg/mL while HM-3A from mushroom *Hypsizygus marmoreus* inhibited U937 cells by 96.2% in 72 h with 100 μg/mL [24,25].

3.10. Hemagglutinating and hemolytic assay

Fig. 4B(a) demonstrated that hemagglutinating activity of HEG-5 at 1000 μg/mL is 16 U. From Table 2 and Fig. 4B(b), HEG-5 has the stable hemagglutinating activity below the temperature of 30 °C. However, HEG-5 has the sensitive thermostability with the decrease of hemagglutinating activity to about 8 U after 40–60 °C treatment, and completely to zero with above 90 °C treatment. The hemagglutinating inhibition experiments showed that all the tested sugar had no inhibition effect concentrations up to 500 mM (Fig. 5), which are similar to those of lectin ORL from the mushroom *Oudemansiella radicata* [28].

The changes of HEG-5 secondary structure after 30–100 °C treatment are shown in Fig. 4C by comparing the circular dichroism profiles. With the increase of treated temperature, the intensities of the positive peak between 195 nm and 200 nm and the negative peak between 205 nm and 220 nm decreased. The negative peak of samples also shifted from 208.6 nm to 218.7 nm after 30–100 °C treatments. During the treatment temperature increase from 30 °C to 100 °C, the percentages of α-helix and β-sheet contents decreased from 13.9% to 7.1%, and 56.5% to 41.8%, respectively, while those of β-turn and random-coil contents increased from 0% to 11.1%, and 29.6% to 40.0%, respectively. Hence, it could be concluded that HEG-5 was a thermo-unstable glycoprotein and its loss of the hemagglutinating activity attributed to the structure disruptions. However, the glycoprotein HEA from dried *H. erinaceus* fruiting bodies had the robust anti-heat property with the stable hemagglutinating activity below 70 °C [17].

Interestingly, HEG-5 had no hemolytic activity by determining the OD_{540} values of hemoglobin released from erythrocytes after adding HEG-5 from 62.5 μg/mL to 10 mg/mL (data not shown).

4. Conclusions

A novel glycoprotein HEG-5 with anti-tumor and hemagglutinating activity was purified from cultured mycelia of *H. erinaceus* CZ-2 by elucidating its structure with GC, FT-IR, ^{1}H NMR and circular dichroism (CD). Its possible 3D structural information was also predicted by MALDI-TOF-MS, NCBI blast search, online NCBInr workspace and online SWISS-MODEL Workspace. An in-depth study of the carbohydrate-binding site, crystal structure and anti-tumor mechanism of HEG-5 is deserved for providing a clue to elucidate its structural–activity relationships.

Acknowledgements

This work was supported by funding from National Natural Science Foundation of China (NSFC 31101269), China Postdoctoral Science Foundation (2013T60648), National Science & Technology Pillar Program during the Twelfth Five-year Plan Period (2012BAD36B05), the Second Class General Financial Grant from the China Postdoctoral Science Foundation (Grant No. 2012M511702), 2012 Excellent Key Young Teachers Project of Jiangsu University, Science & Technology Platform Construction Program (2010DTZ01900) and Priority Science and Technology Innovation Group Project of Jiangxi Province.

References

[1] Zhu Y, Li Q, Mao G, Zou Y, Feng W, Zheng D, et al. Optimization of enzyme-assisted extraction and characterization of polysaccharides from *Hericium erinaceus*. Carbohydr Polym 2014;101:606–13.
[2] Kim SP, Kang MY, Kim JH, Nam SH, Friedman M. Composition and mechanism of antitumor effects of *Hericium erinaceus* mushroom extracts in tumor-bearing mice. J Agric Food Chem 2011;59:9861–9.
[3] Hiwatashi K, Kosaka Y, Suzuki N, Hata K, Mukaiyama T, Sakamoto K, et al. Yamabushitake mushroom (*Hericium erinaceus*) improved lipid metabolism in mice fed a high-fat diet. Biosci Biotechnol Biochem 2010;7(74):1447–51.
[4] Han ZH, Ye JM, Wang GF. Evaluation of *in vivo* antioxidant activity of *Hericium erinaceus* polysaccharides. Int J Biol Macromol 2013;52:66–71.

[5] Mori K, Kikuchi H, Obara Y, Iwashita M, Azumi Y, Kinugasa S, et al. Inhibitory effect of hericenone B from *Hericium erinaceus* on collagen-induced platelet aggregation. Phytomedicine 2010;17:1082–5.
[6] Kim SP, Moon E, Nam SH, Friedman M. *Hericium erinaceus* mushroom extracts protect infected mice against *Salmonella typhimurium*-induced liver damage and mortality by stimulation of innate immune cells. J Agric Food Chem 2012;60:5590–6.
[7] Zhang A, Deng Y, Sun P, Meng X, Zhang J. Structural elucidation of a neutral water-soluble α-D-Glucan from the fungus of *Hericium erinaceus*. J Food Biochem 2011;35:1680–5.
[8] Nagai K, Chiba A, Nishino T, Kubota T, Kawagishi H. Dilinoleoyl-phosphatidylethanolamine from *Hericium erinaceum* protects against ER stress-dependent Neuro2a cell death via protein kinase C pathway. J Nutr Biochem 2006;17:525–30.
[9] Inanaga K. Amycenone, a nootropic found in *Hericium erinaceum*. Pers Med Univ 2012;1:13–7.
[10] Ma BJ, Ma JC, Ruan Y. Hericenone L, a new aromatic compound from the fruiting bodies of *Hericium erinaceums*. Chin J Nat Med 2012;10(5):363–5.
[11] Miyazawa M, Takahashi T, Horibe I, Ishikawa R. Two new aromatic compounds and a new D-arabinitol ester from the mushroom *Hericium erinaceum*. Tetrahedron 2012;68:2007–10.
[12] Zhang A, Sun P, Zhang J, Tang C, Fan J, Shi X, et al. Structural investigation of a novel fuc oglucogalactan isolated from the fruiting bodies of the fungus *Hericium erinaceus*. Food Chem 2007;104:451–6.
[13] Cao J, Nie A, Chen Y, Wang S, Lu H, Yang P. Separation and enrichment of glycoproteins/glycopeptides. Prog Chem 2009;21(9):1888–94.
[14] Cui FJ, Zan XY, Li YH, Yang YY, Sun WJ, Zhou Q, et al. Purification and partial characterization of a novel anti-tumorglycoprotein from cultured mycelia of *Grifola frondosa*. Int J Biol Macromol 2013;62:684–90.
[15] Moradali MF, Mostafavi H, Ghods S, Hedjaroude GA. Immunomodulating and anticancer agents in the realm of macromycetes fungi (macrofungi). Int Immunopharmacol 2007;7:701–24.
[16] Kawagishi H, Mori H, Uno A, Kimura A, Chiba S. A sialic acid-binding lectin from the mushroom *Hericium erinaceum*. FEBS Lett 1994;340:56–8.
[17] Li Y, Zhang G, Ng TB, Wang HX. A novel lectin with antiproliferative and HIV-1 reverse transcriptase inhibitory activities from dried fruiting bodies of the monkey head mushroom *Hericium erinaceum*. J Biomed Biotechnol 2010;2010:1–9.
[18] Huang D, Cui F, Li Y, Zhang Z, Zhao J, Han X. Nutritional requirements for the mycelial biomass and exopolymer production by *Hericium erinaceus* CZ-2. Food Technol Biotechnol 2007;45(4):389–95.
[19] Cui F, Liu Z, Li Y, Ping L, Zhang Z, Lin L. Production of mycelial biomass and exo-polymer by *Hericium erinaceus* CZ-2: optimization of nutrients levels using response surface methodology. Biotechnol Bioprocess E 2010;15: 299–307.
[20] Sluiter A, Hames B, Ruiz R, Scarlata C, Sluiter J, Templeton D. Determination of structural carbohydrates and lignin in biomass laboratory analytical procedure (LAP). Technical Report. USA: National Renewable Energy Laboratory; 2008.
[21] Patrick HKN, Ng TB. A hemolysin from the mushroom *Pleurotus eryngii*. Appl Microbiol Biotechnol 2006;72:1185–91.
[22] Suzuki T, Amano Y, Fujita M, Kobayashi Y, Dohra H, Hirai H. Purification, characterization, and cDNA cloning of a lectin from the mushroom *Pleurocybella porrigens*. Biosci Biotechnol Biochem 2009;73(3):702–9.
[23] Kawagishi H, Suganuma H, Ito K, Mayumi OK, Nagata T. A lectin from an edible mushroom *Pleurotus ostreatus*. Biochim Biophys Acta 2000;1474:209–308.
[24] Chen JN, Wang YT, Wu JSB. A glycoprotein extracted from golden oyster mushroom *Pleurotus citrinopileatus* exhibiting growth inhibitory effect against U937 leukemia cells. J Agric Food Chem 2009;57:6706–11.
[25] Tsai PF, Ma CY, Wu JSB. A novel glycoprotein from mushroom *Hypsizygus marmoreus* (Peck) Bigelow with growth inhibitory effect against human leukaemic U937 cells. Food Chem 2013;141:1252–8.
[26] Cui FJ, Zan XY, Wang HY, Sun WJ, Huang DM, Zhang ZC. Anti-proliferative activity of glycoprotein PLG-3 from the submerged mycelia of *Phellinus linteus* AML1101 in vitro. Adv Mater Res 2012;554–556:1140–7.
[27] Jung EC, Kim KD, Bae CH, Kim JC, Kim DK, Kim HH. A mushroom lectin from ascomycete *Cordyceps militaris*. Biochim Biophys Acta 2007;1770:833–8.
[28] Liu Q, Ng TB, Wang HX. Isolation and characterization of a novel lectin from the wild mushroom *Oudemansiella radicata* (Relhan.: Fr.) Sing. Biotechnol Bioprocess E 2013;18:465–71.

Please cite this article in press as: Cui F-J, et al. Purification and partial characterization of a novel hemagglutinating glycoprotein from the cultured mycelia of *Hericium erinaceus*. Process Biochem (2014), http://dx.doi.org/10.1016/j.procbio.2014.04.008